Harsh Deshpande

B.E. Computer Science | BITS Pilani, Pilani Campus
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Examination	University	Institute	Year	Major	CGPA/%
Graduation	BITS Pilani	BITS Pilani	10/2021-05/2025	Computer Science	8.93/10

WORK EXPERIENCE & INTERNSHIPS

- **Intern** | ***Capital One*** *(Jan'25-Present)*
  - Intern in the Customer Solutions Team. Gained experience to SQL and was responsible for refactoring SQL scripts for the data pipeline.
  - **Tech stack: Python, Snowflake**
- **Summer Intern** | ***Eightfold.ai*** *(May'24-Jul'24)*
  - Intern in the **Data Platform Team**.Worked on File Ingestion as part of a summer internship
  - Developed and implemented an end-to-end ingestion solution for specific entities in the database schema. Generated detail reports in the parsing of the files, ensuring comprehensive data analysis. Developed and executed unit tests to validate the ingestion and transformation process.
  - Worked on creating the file ingestion page, implementing front-end changes as well. Added debugging tools to the same page which enhanced real-time error detection and troubleshooting.
  - Refactored code and deprecated fields,improving maintainability and performance. Conducted configuration migrations for notification settings to support new ingestion workflows and system updates
  - **Tech stack: Python, React, Flask, AWS S3**

PUBLICATIONS

1. **CultureShift: Mapping Temporal Cultural Evolution in Vision-Language Models** [PDF]
   Gautam Jajoo, **Harsh Deshpande**, Hamna, Pranjal A Chitale
   VLMs-4-All Workshop, Conference on Computer Vision and Pattern Recognition (CVPR '25)

RESEARCH PROJECTS

- **Multimodal Learning for Knowledge Based Visual Question Answering**
  *(Guide: **Prof. Poonam Goyal**)* *(Jan'24-Dec'24)*
  - Tasked with implementing the architecture based on the REVIVE model (Regional Visual Representation Matters in Knowledge Based Visual Question Answering)
  - Found the results on a self-curated OK-VQA like dataset which was pertinent to advertisements.Assisted in obtaining the results for the paper to be published and comparing them with other SOTA models
  - Learnt about **Regional Visual Representations and Multimodal Learning** in general. Specifically spent time learning about **GLIP,CLIP as well as various Image Captioning models**. Studied the **REVIVE,KAT and PiCA** architectures.
- **Contrastive Learning Methods on Point Cloud Data**
  *(Guide: **Prof. Vinay Chamola**)* *(May'23-Jul'23)*
  - Studied various self-supervised methods including **Contrastive learning techniques such as SimCLR, SimCLR v2**, as well as other representation learning methods such as **BYOL** for the use case of object detection by UAVs using 3D LIDAR data in low-light conditions

TECHNICAL SKILLS

- **Programming Languages**: C, C++, Java, Python, Prolog, JavaScript
- **Frameworks & Tools**:
  Proficient: PyTorch **Familiar**: TensorFlow, CUDA, Pandas, JAX
- **Database Technologies**: SQL (MySQL, PostgreSQL), NoSQL (MongoDB)
- **Version Control & DevOps**: Git, GitHub, Docker, CI/CD pipelines

RELEVANT COURSEWORK

- **Computer Science** Natural Language Processing, Artificial Intelligence, Operating Systems, Discrete Structures for Computer Science , Data Structures and Algorithms , Logic in Computer Science , Theory of Computation , Design and Analysis of Algorithms, Computer Architecture
- **Mathematics** Graph Theory , Game Theory , Multivariate Calculus, Probability and Statistics, Linear Algebra and Complex Analysis

RELEVANT COURSE PROJECTS

- **Automated Pipeline to Generate Posters for Welfare Schemes**,
  (Natural Language Processing Course Project) Code *(Sep'24-Nov'24)*
  - Designed and implemented a pipeline to generate posters for a welfare scheme given a thorough description of the welfare scheme.
  - Implemented the PRISM method within the pipeline as introduced in the paper He, Y., Robey, A., Murata, N., Jiang, Y., Williams, J., Pappas, G. J., Hassani, H., Mitsufuji, Y., Salakhutdinov, R., & Kolter, J. Z. (2024). **Automated black-box prompt engineering for personalized text-to-image generation.**
  - The pipeline consisted the following stages: (a) Generation of topic-specific write-ups given the description of the welfare scheme, (b) Generation of character context to ensure consistency of characters between posters of the same welfare scheme, (c) PRISM method to enhance the image generation prompt.
  - Ensured that the entire pipeline was **interpretable** and helpful for a human-in-the loop to **mitigate bias** in the generated posters. This was done by generating interpretable logs and character contexts which could be read by a human in the loop. Any bias could be eliminated on the basis of this.
  - Observed an increase in CLIP score from **0.626 to 0.677** post integrating the PRISM method to the pipeline.

- **Compiler for a Toy Language**, *(Compiler Construction Course Project) Code* *(Apr'24-May'24)*
  - Implemented the front-end of a compiler, from scratch using only C, which supports static arrays,arithmetic and boolean operations,block scoping,input/output statements, and declarative, conditional,iterative and function call statements.
  - Includes a DFA based lexer and a top-down LL(1) predictive parser with panic mode error recovery

- **Hilbert R Tree Implementation**, *(Data Structures and Algorithms Course Project) Code* *(Apr'23-May'23)*
  - C Implementation of the Hilbert R Tree Data Structure:Designed an efficient implementation of the Hilbert R Tree, a spatial access method that improves upon the standard R-Tree by using Hilbert space-filling curves to better preserve spatial locality.

ACADEMIC ACHIEVEMENTS

- **Kishore Vaigyanik Protsahan Yojana (SA) Fellow** | ***Issued by IISc Bangalore*** *(2019)*
  - All India Rank 623
- **Kishore Vaigyanik Protsahan Yojana (SX) Fellow** | ***Issued by IISc Bangalore*** *(2020)*
  - All India Rank 1680

TEACHING ASSISTANT POSITIONS

- **Teaching Assistant** for the course, **CSF222 : Discrete Structures for Computer Science**,
  Led weekly tutorial sessions to reinforce and elaborate on the topics covered in the lectures. Facilitated group discussions and collaborative problem. Designed and prepared comprehensive tutorial sheets that aligned with the course curriculum. *(Aug'23-Dec'23)*
- **Teaching Assistant** for the course, **CSF301 : Principles of Programming Languages**
  Currently working with Prof. Dhruv Kumar to create bots to assist on instructing as well as evaluating students in the Principles of Programming Languages course. We are working towards understanding the impact as well as limitations of LLMs in assisting the students' process of learning in the course. *(Aug'24-Dec'24)*